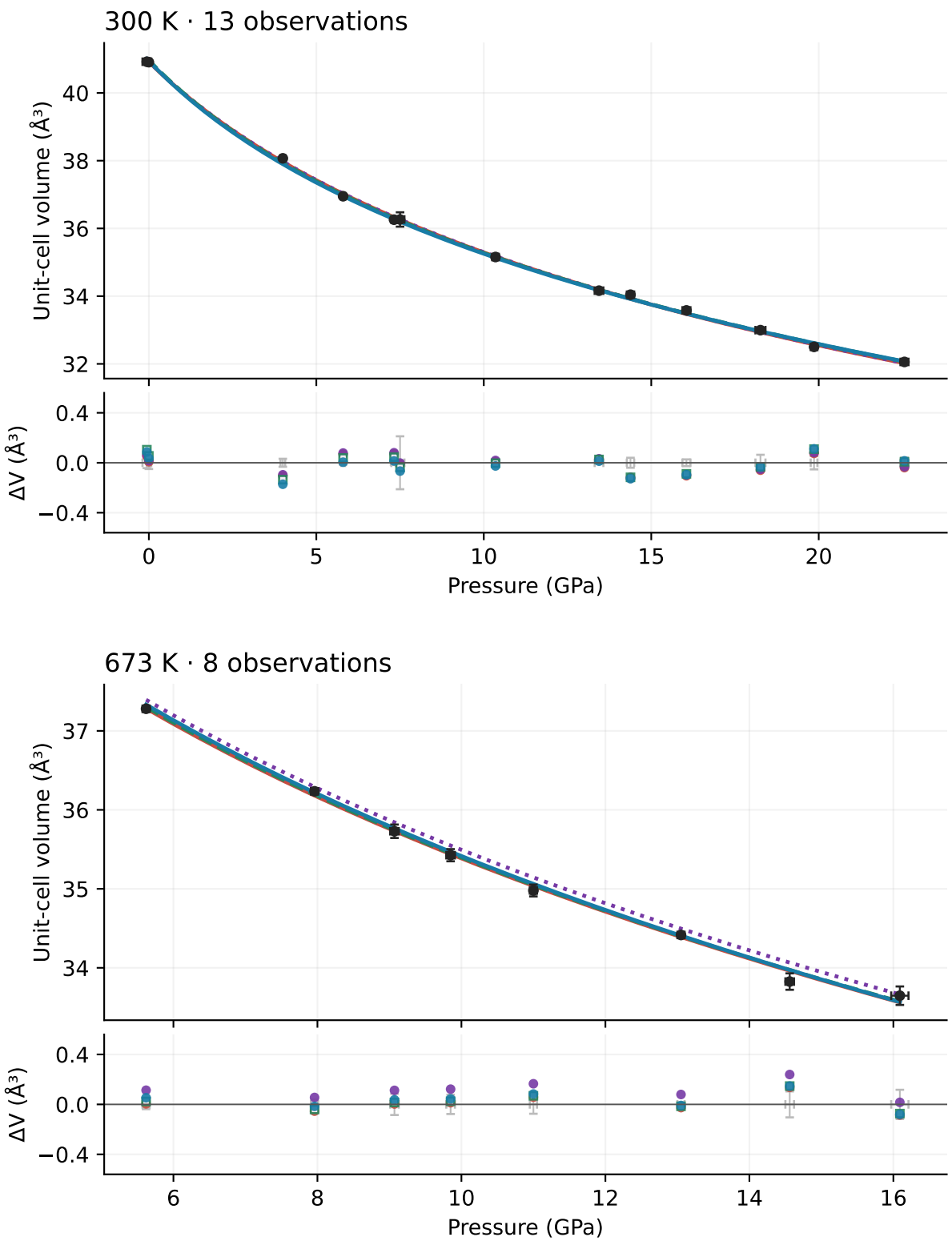
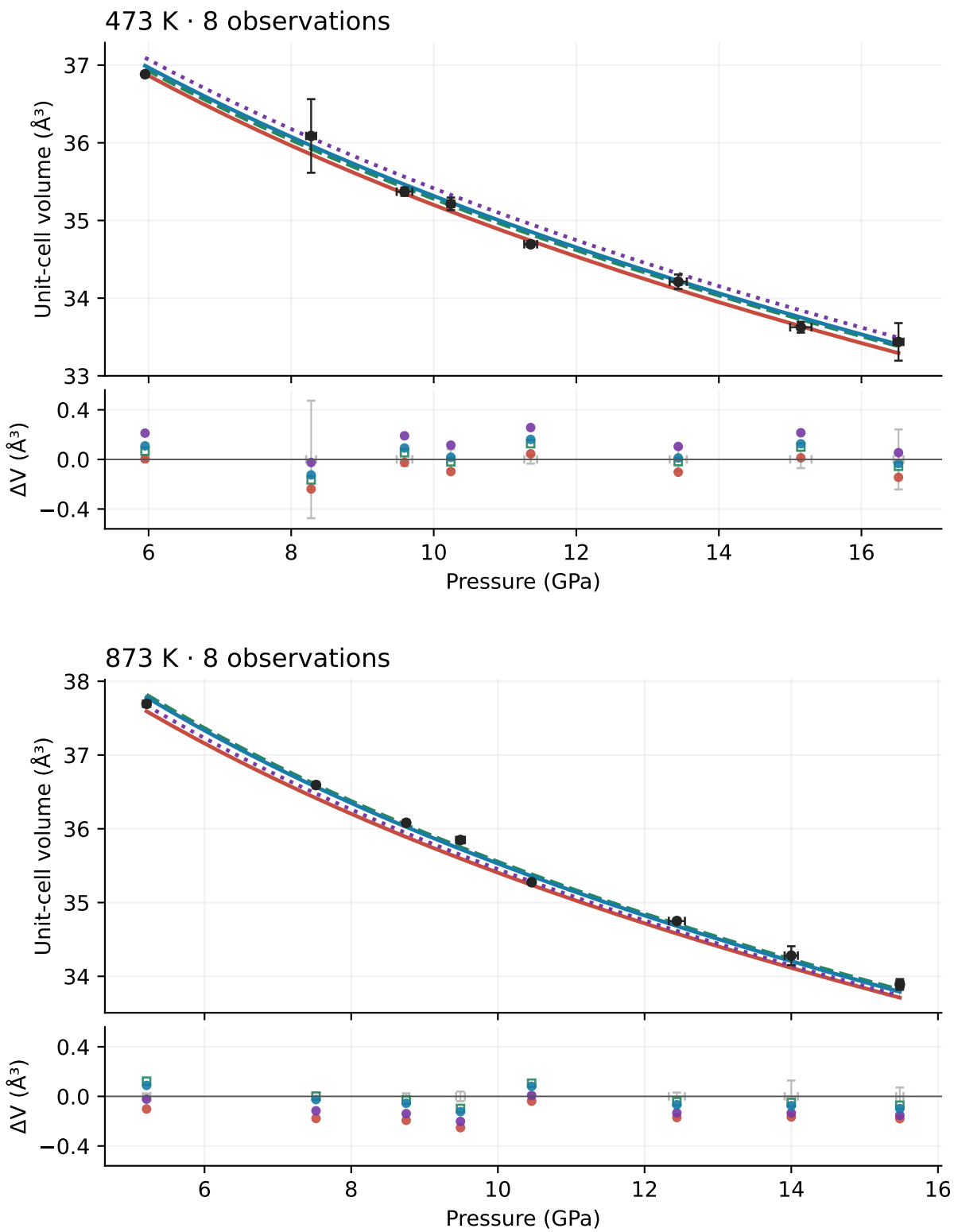
Brucite: does a joint BM3–MGD fit improve the agreement?

Joint BM3–MGD refit · 6 free parameters
Joint empirical thermal refit · 7 free parameters
Current continuous model · ambient expansion refit + published compression parameters
Original published isotherms · printed expansion and tabulated moduli





Both joint fits minimize unweighted volume residuals to the same 37 P–V–T + 11 ambient T–V rows.
ΔV = calculated − observed volume at observed pressure. Grey residual bars show reported V and P errors about zero.
MGD uses Tr = 300 K, n = 5 atoms per $Mg(OH)_2$ formula unit, integrated γ–Θ law and reference-subtracted Debye energy.
In-sample diagnostic fits: no phase-equilibrium, calorimetric or pressure-scale validation; catalog records unchanged.